

“Dear Professor Lukoff”: GenAI and the Collapse of Costly Signals in Communication

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Abstract

Personalized messages once carried proof of their own effort: a cold email that engaged with a professor’s publications showed that a student had found them, read them, and cared enough to write. GenAI severs that link. Drawing on signaling theory, I argue that GenAI converts assessment signals, which cannot be produced without the ability and effort they advertise, into conventional signals that cost nothing to fake. I distinguish three jobs a message can do (transmission, ritual, and assessment) and show that GenAI often leaves the first intact while gutting the other two. The resulting harm is a commons problem with a regressive distributional consequence: when gatekeepers stop trusting open channels, they retreat to personal networks, and opportunity shifts toward the already-connected. I outline design responses that restore costly signals by choice rather than by mandate, and policy questions about the obligations of platforms and institutions to preserve open channels.

Keywords

AI-mediated communication, generative AI, signaling theory, trust, costly signals

1 Introduction

A local high school student recently emailed me to request a summer internship in my research lab. The email, addressed “Dear Professor Lukoff,” explained why my work interested the student, mentioned three of my publications, and connected them to the student’s own background. From those cues I would once have inferred a student who can find and read research papers, who spent an hour or two with mine in particular, and who has a genuine interest in my research area. Three years ago I would have been impressed, and might have invited them in.

Today I hesitate, and often set such emails aside. Anyone can prompt an agentic AI tool to research all of the computer science professors in the San Francisco Bay Area and draft a personalized internship request for each one. Three years ago I received a handful of these emails per year; now I receive over a hundred. Some show clear signs of AI writing. With many I cannot tell; a representative example, anonymized, appears in Appendix A. A few students surely do the research and writing themselves, and they now suffer from a devalued channel that no longer carries a signal of value.

This paper takes a position in four parts. First, signaling theory explains what has broken: GenAI converts assessment signals into conventional ones. Second, whether that conversion matters depends on which of three jobs a message is doing: transmitting information, performing a ritual, or serving as an assessment. Third, the resulting harm is a commons problem whose costs fall unevenly,

punishing honest senders and people without networks while eroding gatekeepers’ ability to find talented strangers. Fourth, I weigh candidate responses and argue that redesigning channels so senders can voluntarily take on verifiable costs holds more promise than disclosure mandates, which no one can enforce.

2 Background: Signals in Mediated Communication

AI-mediated communication describes settings where a computational agent modifies, augments, or generates messages on behalf of a communicator [5]. Research in this area has documented effects within a communication session: receivers rate profiles as less trustworthy when they suspect AI authorship [8], and AI involvement shifts blame and trust between partners [6]. This paper concerns a slower effect that spans channels rather than sessions: the erosion of what messages can prove about their senders. This is quickly becoming true across almost all of our communication channels, whether we like it or not.

Signaling theory distinguishes two kinds of signals [12, 14]. Assessment signals are reliable because a sender cannot produce them without the quality they advertise: in Spence’s classic example, a university degree signals ability to employers because less able students would need far more time and effort to complete one. Conventional signals have no built-in connection to the quality they advertise; they stand for it by shared convention, and stay reliable only while faking them carries a cost. A resume line claiming “strong research skills” is conventional: anyone can type it, and it stays trustworthy only as long as a lie is likely to be caught. Donath [3] argued that online self-presentation runs mostly on conventional signals, kept honest by the social cost of exposure, and warranting theory holds that receivers discount cues in proportion to how cheaply a sender could fake them [13]. When the cues in the message itself become cheap, receivers fall back on weak proxies such as fit and phrasing, and honest senders with atypical profiles are exactly who gets misclassified (see Appendix A for how this plays out in my own reading of one representative email).

The personalized cold email was an assessment signal: before GenAI, a student could not write sensibly about my papers without reading them. Now it is a conventional signal that costs almost nothing to produce. Writing the email now costs so little that it reveals nothing about the sender’s effort or ability. Students did not become more deceptive; the link broke between the cost of the signal and the qualities it once certified.

3 Three Jobs a Message Can Do

Whether that break matters depends on what a message is for. A message can do at least three jobs (Table 1).

Table 1: Three jobs a message can do, and what GenAI does to each.

Job	The message exists to...	Examples	What GenAI does	What GenAI does at scale
Transmission	Convey information clearly	Policy announcement, documentation	Often a gain if used skillfully: clarity, translation, accessibility	Low-cost content floods the channel faster than receivers can read
Ritual	Sustain a relationship through the act itself	Birthday card, condolence note	Corrosive: keeps the words, deletes the effort that was the content	Mass-produced sincerity reads as none
Assessment	Serve as evidence about the sender	Exam essay, cover letter, cold email	Breaks the instrument, whatever the sender intended	One prompt “personalizes” hundreds of applications

Scale is the multiplier across all three rows. Personalization was self-limiting when each message cost time; GenAI removes that ceiling, and one prompt can now tailor a message to every professor in a region or every open role on a job board. “Personalized” outreach begins to behave like spam.

In transmission, clarity is the point and the sender’s effort is beside it; AI assistance often helps, including for non-native speakers. In ritual communication, the effort is the message. Carey [1] called this the ritual view of communication, in which the act of communicating itself sustains the relationship; a generated card keeps the words and loses the content. People sense this: even those open to AI help with hard conversations, such as breakup messages, worry that delegating the words hollows out the act [4]. In assessment, the artifact is evidence about its sender, and GenAI breaks that instrument regardless of intent.

The internship email did all three jobs at once: it transmitted a request, performed interest in my work, and demonstrated research ability. GenAI leaves the first job intact, since the prose in these emails is clear and often polished. It guts the other two: the same email no longer certifies interest or ability.

4 A Signaling Commons

The erosion of this costly signal is a harm that accumulates across many small choices. A student’s choice to use AI to reach out to dozens of faculty at once. My choice to ignore those emails. Together they degrade a signaling commons, and the students who did the work themselves get punished along with everyone else. No single actor behaves badly, and no single deployment decision caused the damage, which places this squarely in the class of risks this workshop targets: gradual, distributed, and hard to attribute.

There is a longer-run cost too: reading papers closely is one way a student starts to become a researcher, and if the payoff for that effort disappears, so may the practice. Researchers are only beginning to measure what LLM use does to this kind of critical engagement [15].

5 Who Absorbs the Loss

When I stopped trusting cold emails, I did not stop screening applicants. I would like to find a better alternative, but so far I have largely turned to a substitute: recommendations from people I know. That substitution helps me manage the deluge, but it is potentially regressive in aggregate. The cold email was an open channel for a

high schooler with no network, a first-generation college student, or an applicant from an unranked program. As I weight “who you know” more heavily, opportunity moves toward people with more existing social capital.

Hiring may be following the same pattern. Applications per hire have roughly tripled since 2021 to more than 300 per open role [2], with AI-tailored resumes and cover letters driving much of the flood, and employers now lean on referrals and even “backdoor references” from mutual connections. Candidates with thin networks face a diminished channel. Organizations lose something too: the capacity to spot a talented stranger who fits the role. Social capital compounds and the well-connected pull further ahead. No one decided on this second-order effect of generative AI, and no single actor can be held to account for it.

I suspect the same substitution is underway wherever assessment channels are collapsing, that is, wherever a written artifact once served as evidence about its sender: admissions essays, scholarship applications, and graduate school applications. The pattern reaches beyond academia. On dating apps and home exchange platforms, a personalized first message once proved that a stranger had read your profile and meant it. On freelance platforms, a tailored bid once showed that a worker had studied the project brief. In civic life, congressional offices have long treated an individually written letter as a stronger signal of constituent concern than a form letter. Receivers in all of these channels now face the same doubt. I hope to explore this suspicion in our workshop discussions.

6 Design: Restoring Costly Signals by Choice

My research examines how interface design shapes attention and agency, including a taxonomy of attention capture deceptive designs [11] and “adaptable commitment interfaces” that let users voluntarily bind their future behavior [10]. Commitment devices are voluntary costly signals, and they suggest a design direction for collapsed channels: let senders bind themselves. Candidate mechanisms include rate-limited applications (scarcity makes each one meaningful again), a named endorser or a refundable deposit (staking), and a short live conversation (synchronous verification). Process provenance, evidence of how a message was made, may matter more than whether AI touched it. The design question is whether such mechanisms can restore assessment value without recreating the exclusion that open channels once bypassed: a deposit is a costly signal, and also a means test. Alternative formats shift the

cost rather than settle the question. Asking applicants to submit a short video restores a time cost and puts a person behind the words, but it invites bias on appearance, accent, and production quality, and improving AI video generation will erode even that floor.

7 Policy: Beyond Disclosure Mandates

Disclosure mandates are the most commonly proposed response, and I am skeptical of them here: no one can verify them, and they assign enforcement to the party with the least incentive to comply. Freelance work already shows the pattern: workers tend to disclose AI use only when asked, while clients overestimate how well they can detect it [7]. Online communities that have tried blunt AI bans face the same verification problem [9].

The harder policy questions concern infrastructure and obligation. Which verified-effort primitives should platforms be required to support? Do institutions that benefit from open channels, universities, employers, funding agencies, owe anything toward maintaining them? And who bears the residual cost when signals stay collapsed: the senders who lose a channel, the receivers who drown in volume, or the platforms that control and often profit from the traffic?

8 Questions for This Workshop

- How do we measure a channel's signal value after GenAI adoption? Which signals are collapsing today?
- Can design restore costly signals without rebuilding the exclusion that cold channels once bypassed?
- How should norms differ across transmission, ritual, and assessment communication, and who legitimately sets them?
- What obligations, if any, do platforms and institutions have to preserve open channels through which talented strangers can credibly signal ability and interest?
- How can gatekeepers cope with AI-polished volume without retreating to their networks, a substitution that rewards existing social capital and compounds rich-get-richer dynamics?

9 Conclusion

GenAI has not necessarily made communicators dishonest; it has made honesty indistinguishable from its imitation in any channel where the message once certified its sender. The costs of that shift are diffuse, cumulative, and regressive, falling hardest on people whose only capital was the willingness to do the work. Rebuilding channels where effort can still show, without walling them off to outsiders, is a research, design, and policy problem that this workshop can explore.

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A A Representative Inquiry

The email below is one of the more than one hundred research inquiries I received in the past year. To protect the sender, a high school student, I have anonymized it and gently rephrased the wording: the name, school, and location are removed or generalized, and the wording is lightly altered so that the sender's own words are not published verbatim. The structure, register, and level of specificity are preserved. I suspect this email was drafted in large part by AI: the prose carries familiar signs of AI writing, and the targeting is a stretch, since the student's stated background in applied machine learning connects only loosely to my lab's HCI research. But I cannot be certain. That is the receiver's predicament this paper describes: I am left inferring from weak cues like fit and phrasing, and a student who wrote every word themselves would have no way to prove it.

Subject: Summer 2026 Research: High school junior with applied ML experience hoping to contribute to digital wellbeing research

Dear Professor Lukoff,
My name is [redacted], and I am a high school junior in the San Francisco Bay Area. I am reaching out because I came across your recent work on digital attention heuristics and intentional technology use. I am drawn to research that combines measurement, evaluation, and practical usability, and your work stood out to me for that reason. My background is in applied machine learning and reproducible evaluation. Alongside independent ML projects, I

have experience organizing experiments, working with noisy datasets, and building analysis pipelines in Python. If there might be a way for a high school student to contribute to your group this summer, I would be glad to help implement analysis pipelines, replicate metrics, and clean and aggregate study datasets. If it would be easier, I would also be grateful for a brief 10–15 minute call with you or

a member of your group to see whether there could be a fit. If not, I would appreciate any guidance on whether a graduate student in your group might be open to a smaller, well-scoped project.

Best regards,

[redacted]

High School Junior, [redacted] High School, CA